

Additional TensorFVM analytic benchmarks

Three-dimensional ABC / Beltrami flow and advected shear wave

1. Problem and exact solution

ABC $u=(\sin z+\cos y, \sin x+\cos z, \sin y+\cos x) \exp(-\nu t)$.

$p=-\rho |u|^2/2 + \text{mean gauge}$. Curl $u=u$; Laplacian $u=-u$.

The nonlinear term is a gradient and balances pressure.

Shear: $u=\sin(y-0.7t) \exp(-\nu t)$, $v=0.7$, $w=0$, $p=0$.

This checks transport phase with nonzero mean momentum.

2. Methods

Existing production face-MAC midpoint solver, full symmetric stress, shared momentum flux and compatible FFT projection.

Velocity at end time on real faces; pressure at last midpoint.

ABC: $n=24/32/48$, $dt=0.005$, $end=0.1$ s, true 3D.

Shear: $n=16/24/32$, four z layers, $dt=0.01$, $end=0.5$ s.

Physical errors $<3\%$; all-step conservation/nonlinear gates.

3. Reproducibility

Raw accepted states, CSV curves, source hashes and independent NumPy reconstruction accompany each run. No production solver changes.

Results and scope

abc	24x24x24	u	L2=0.005699%	pass=True
abc	32x32x32	u	L2=0.003209%	pass=True
abc	48x48x48	u	L2=0.001427%	pass=True
advected-shear	16x16x4	u	L2=0.620363%	pass=True
advected-shear	24x24x4	u	L2=0.276903%	pass=True
advected-shear	32x32x4	u	L2=0.156024%	pass=True
abc	16x16x16	u	L2=0.012786%	pass=False
advected-shear-lbm	32x32	u	L2=0.230176%	pass=True
advected-shear-lbm	64x64	u	L2=0.054381%	pass=True

Observed orders:

abc-velocity_l2: [1.9965791689542742, 1.998281267117936]

abc-pressure_l2: [1.9914511555969971, 1.9956845638247966]

advected-shear-shear_velocity_l2: [1.9894089548853335, 1.9940800618877303]

ABC n16 pressure failure retained; excluded from formal qualification.

D2Q9 cannot compare with three-dimensional ABC.

Shear pressure error is absolute / initial dynamic pressure, since $p=0$.

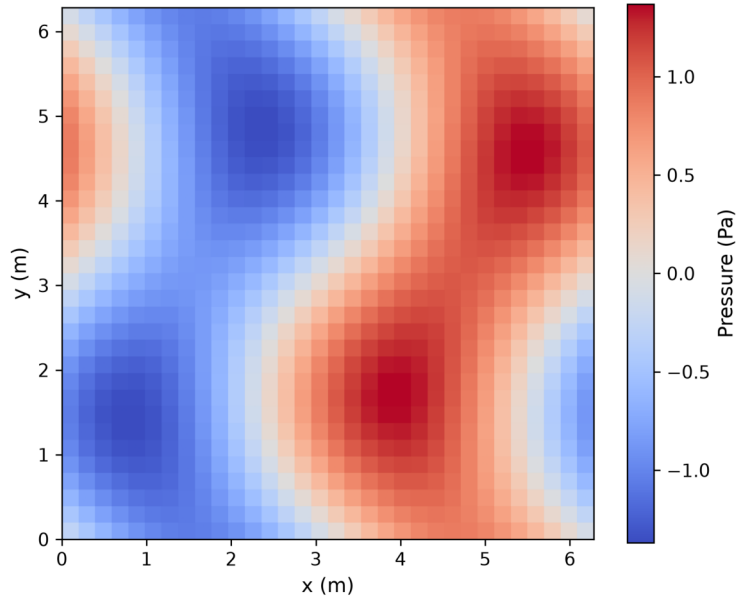
LBM production BGK uses float64 populations, upstream float32 weights.

No generalized speed, memory or industrial qualification claim.

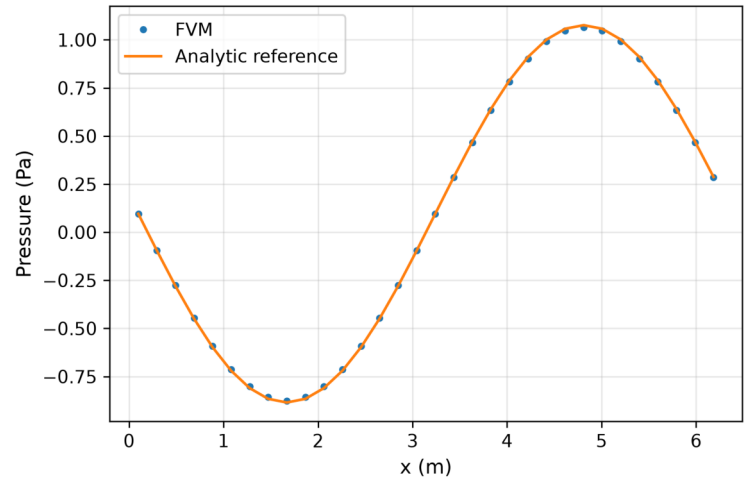
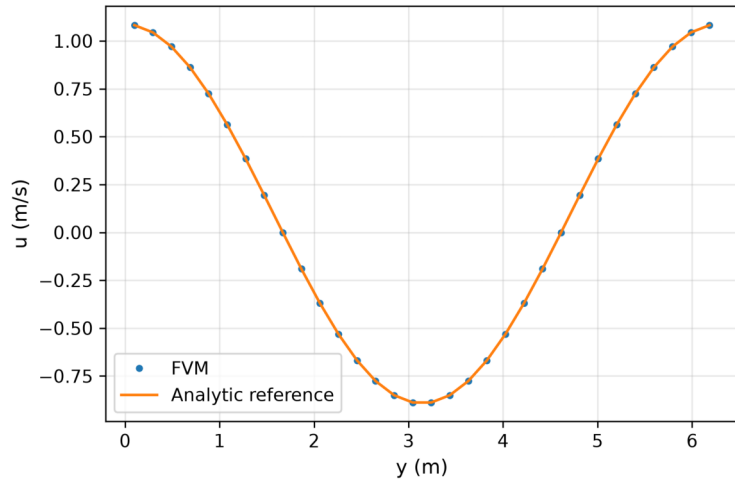
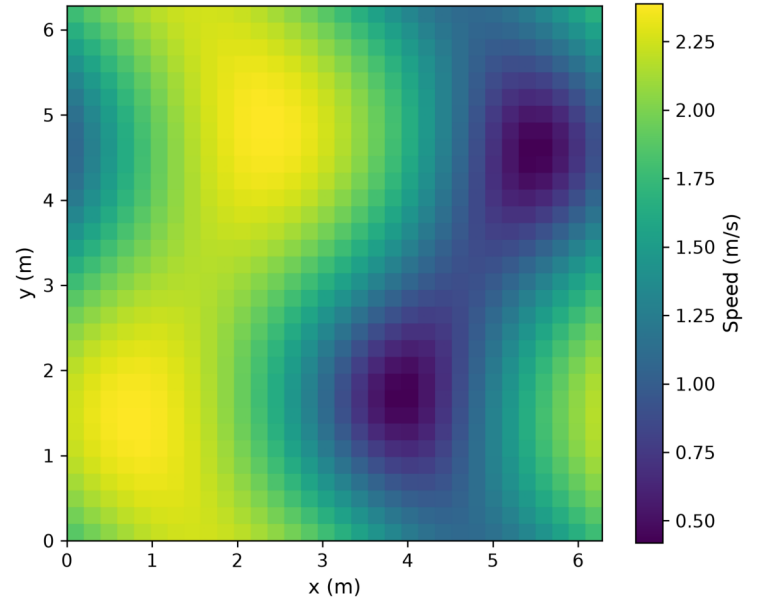
Research manuscript; not peer-reviewed.

abc 32x32x32

abc 32x32x32: first z cell slice

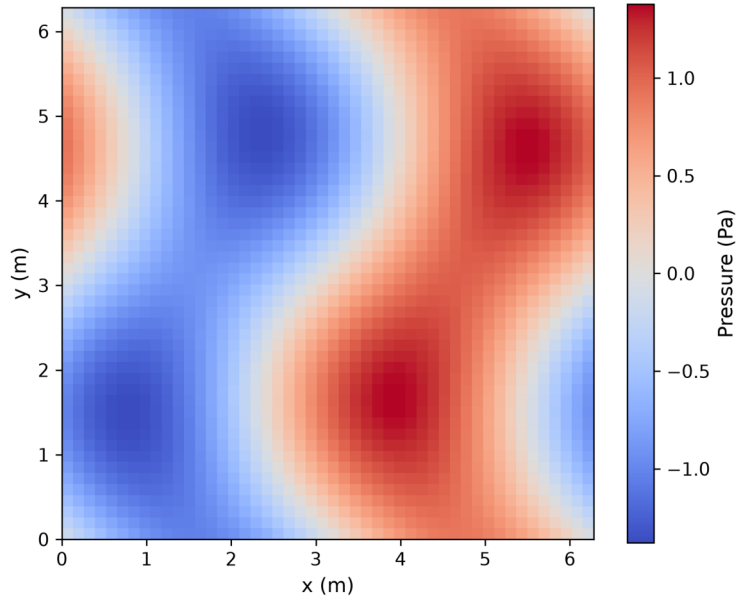


abc 32x32x32: first z cell slice

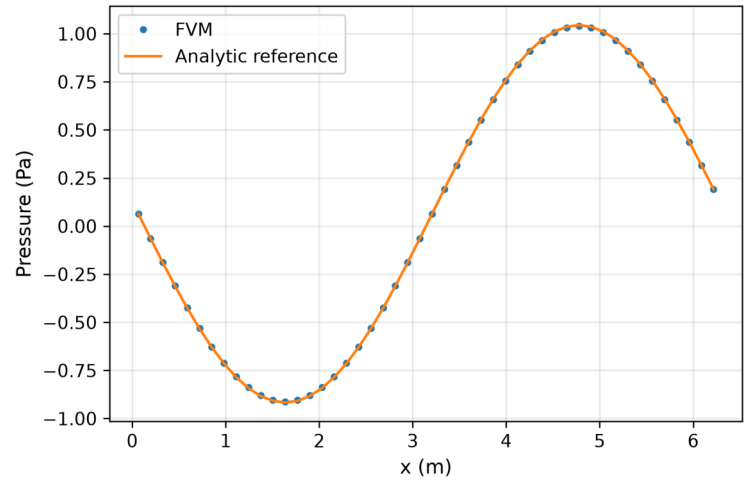
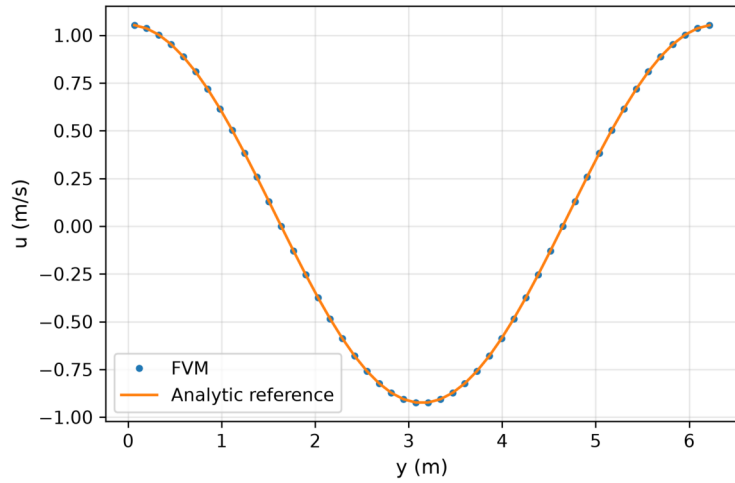
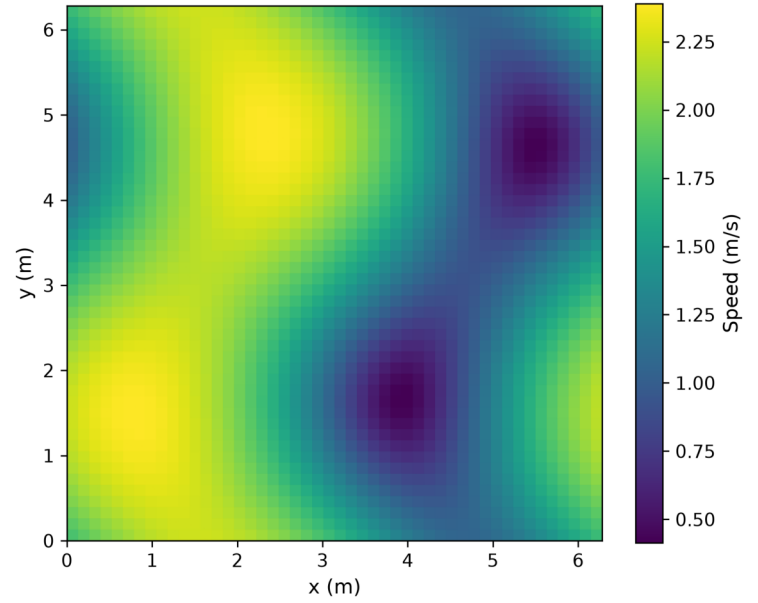


abc 48x48x48

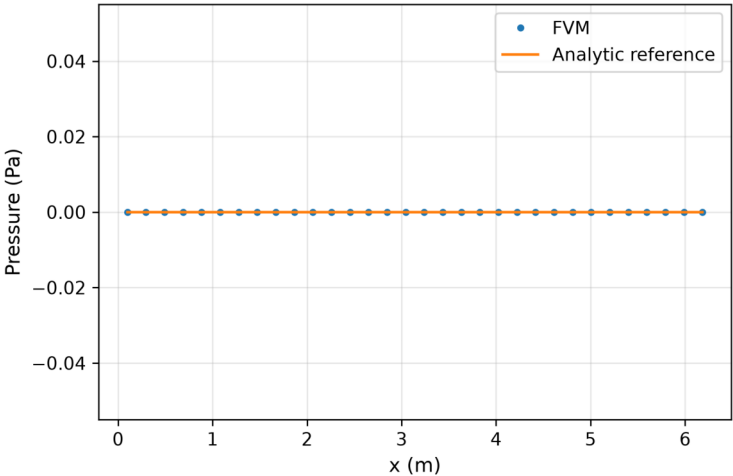
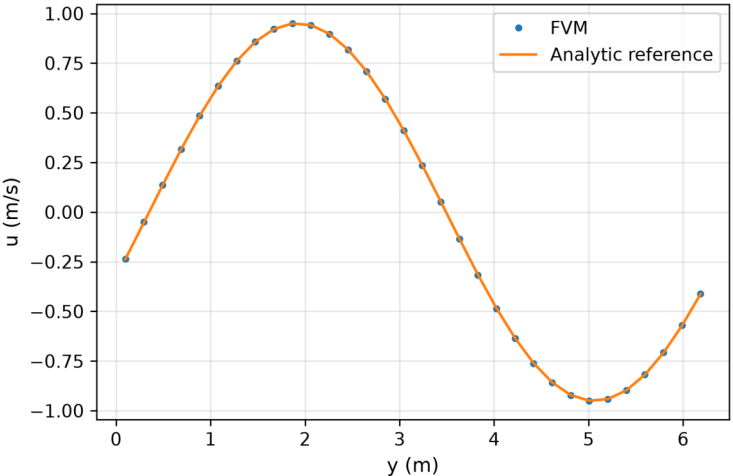
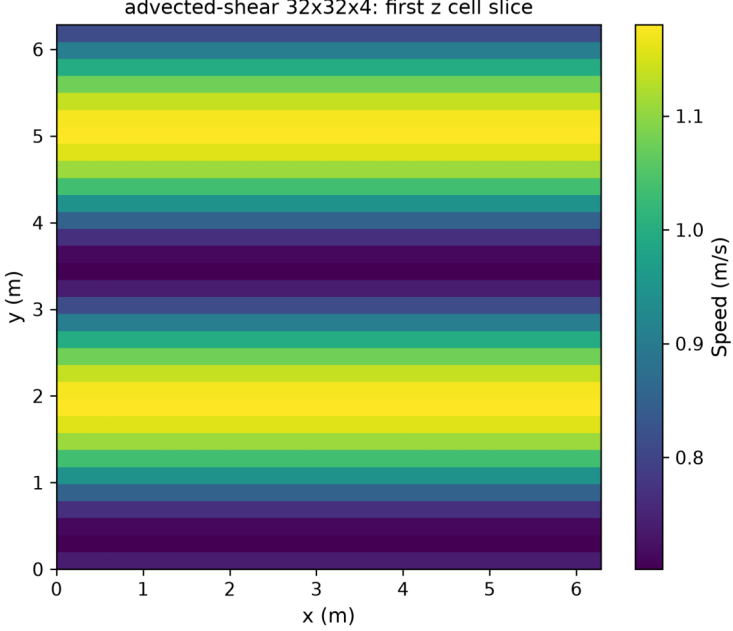
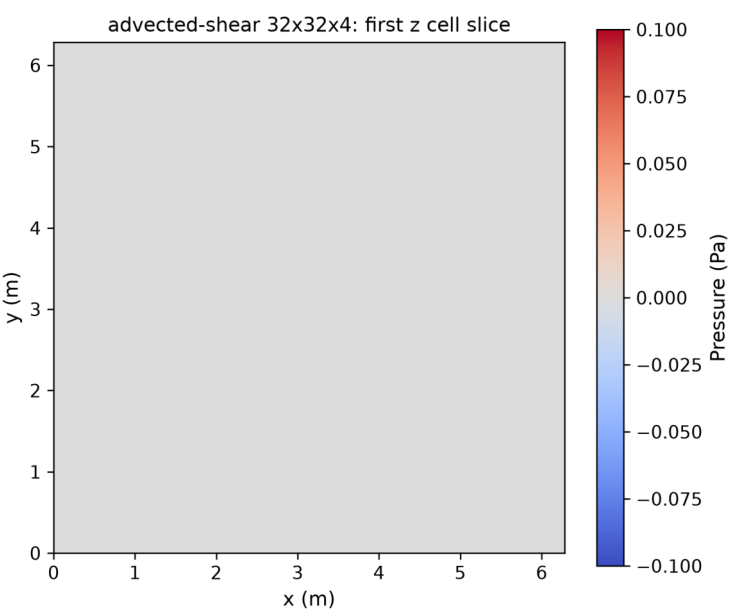
abc 48x48x48: first z cell slice



abc 48x48x48: first z cell slice



advected-shear 32x32x4



advected-shear-lbm 32x32

